

Vishal Kumar Allam

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PROFESSIONAL SUMMARY

- Forward-Deployed and AI Engineer with **6+ years of experience** designing and delivering enterprise software, Generative AI applications, data platforms, cloud-native services, APIs, and client-facing technical solutions across healthcare, insurance, and enterprise technology environments.
- Over **2+ years of hands-on experience** developing and deploying **AI, ML, Generative AI, LLM, Retrieval-Augmented Generation (RAG), agentic AI, and intelligent workflow solutions** using Python and modern AI engineering frameworks.
- Experienced in transforming ambiguous business challenges into **working demos, prototypes, proofs of concept, proof-of-value solutions, and MVPs** that demonstrate technical feasibility and create clear paths toward enterprise-scale implementation.
- Strong hands-on development experience with **Python, JavaScript, TypeScript, FastAPI, Flask, React, LangChain, LlamaIndex, PyTorch, REST APIs, microservices, and SQL** for building scalable AI-enabled applications and backend services.
- Experienced developing **agentic AI solutions** that combine Large Language Models, tool calling, enterprise APIs, structured and unstructured data, workflow orchestration, retrieval systems, business logic, and human-in-the-loop controls.
- Hands-on experience integrating AI solutions with leading model ecosystems including **OpenAI, Anthropic Claude, Google Gemini, and Mistral**, selecting model and architecture patterns based on business requirements, latency, quality, security, and cost.
- Strong experience with **Retrieval-Augmented Generation**, including embeddings, semantic search, vector databases, document ingestion, chunking, metadata filtering, retrieval pipelines, prompt engineering, contextual grounding, and response generation.
- Experienced working directly with **business stakeholders, product teams, architects, engineering teams, and client-facing groups** to conduct technical discovery, validate requirements, communicate architecture decisions, and rapidly iterate on working solutions.
- Hands-on cloud engineering experience across **AWS, Azure, and GCP**, including serverless computing, storage, compute, IAM, API services, networking concepts, cloud-native deployment, Terraform, and Infrastructure as Code.
- Experienced with enterprise data platforms and engineering technologies including **Palantir Foundry, Databricks, Snowflake, SQL, PySpark, ETL/ELT, data pipelines, data modeling, CDC, batch processing, and API-based data integration**.
- Strong knowledge of **MLOps and LLMOps** practices including model deployment, prompt management, evaluation frameworks, experiment tracking, model and agent monitoring, versioning, observability, CI/CD, and production AI operations.
- Experienced building enterprise AI solutions with consideration for **Responsible AI, security, privacy, data handling, authentication, authorization, governance, model risk, scalability, latency, and operational requirements**.
- Proven ability to work independently in **fast-paced and ambiguous client-facing environments**, owning technical build work from discovery and architecture through implementation, demonstration, documentation, deployment, and engineering handoff.
- Strong communicator experienced in **technical deep dives, executive demonstrations, architecture discussions, solution workshops, stakeholder presentations, technical documentation, reusable assets, demo scripts, setup guides, and implementation runbooks**.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, JavaScript, TypeScript, C#, Java, SQL, FastAPI, Flask, React, Pydantic, RESTful Services, Microservices, Object-Oriented Programming.
- **Generative AI / Agentic AI:** GenAI, LLMs, RAG, AI Agents, Agentic Workflows, OpenAI, Anthropic Claude, Mistral, Google Gemini, LangChain, LlamaIndex, Prompt Engineering, Tool Calling, Human-in-the-Loop.
- **AI/ML & Data:** PyTorch, TensorFlow, Embeddings, Vector Databases, Semantic Search, Model Evaluation, Data Science, PySpark, Spark, ETL/ELT, Data Pipelines, Data Modeling, CDC, Databricks, Snowflake, Palantir Foundry/AIP.
- **Cloud & Infrastructure:** AWS, Azure, GCP, AWS Lambda, S3, IAM, CloudWatch, Azure OpenAI, Serverless Computing, Docker, Terraform, Infrastructure as Code, Cloud Storage, Compute, Networking.
- **APIs & Enterprise Integration:** REST APIs, FastAPI, Flask, Swagger/OpenAPI, JSON, API Gateways, Microservices, Event-Driven Architecture, OAuth2/OIDC, JWT, SAML, RBAC, Enterprise System Integration.
- **MLOps / LLMOps & DevOps:** Model Deployment, Prompt Management, Model Monitoring, Agent Evaluation, LLM Observability, CI/CD, Azure DevOps, Jenkins, TeamCity, GitLab CI/CD, GitHub Actions, Git, Docker.
- **Solution Engineering:** Demos, Prototypes, POCs, MVPs, Proof-of-Value, Technical Discovery, Executive Demos, Technical Deep Dives, Solution Architecture, Responsible AI, Security, Privacy, Technical Documentation, Runbooks, Reusable Assets.

EXPERIENCE

Software Engineer AI *PCMI LLC (Full Time)*

Oct 2024 – Present
Dallas, TX, USA

- Design and deliver client-facing **AI, GenAI, and Palantir-based demos, prototypes, POCs, and MVPs** using Palantir Foundry/AIP, Python, FastAPI, SQL, LLMs, REST APIs, and cloud services for enterprise insurance and operational use cases.
- Translate ambiguous client and business problems into working technical concepts using **models, AI agents, RAG, data pipelines, workflows, APIs, integrations, and user experience patterns**, collaborating with stakeholders and Solution Architects.
- Develop agentic AI and RAG solutions using **OpenAI, Anthropic Claude, Google Gemini, Mistral, LangChain, embeddings, vector databases, semantic retrieval, prompt engineering, and human-in-the-loop controls**.
- Build **Python/FastAPI services, Palantir Foundry pipelines, PySpark/SQL transformations, Ontology-driven applications, reusable APIs, and enterprise integrations** connecting AI models with governed business data and operational systems.
- Validate **technical feasibility, data dependencies, integration complexity, architecture assumptions, security, privacy, scalability, model quality, latency, and operational requirements** before moving prototypes toward production delivery.
- Implement secure cloud and enterprise integration patterns using **AWS, Azure, Terraform, OAuth2/OIDC, JWT, SAML, RBAC, Docker, CI/CD, logging, monitoring, and LLMOps/MLOps** practices.
- Package solutions into reusable assets including **Python libraries, prompt templates, agent patterns, API connectors, demo scripts, setup guides, technical collateral, runbooks, and engineering handoff materials** for broader team reuse.

Environments: Palantir Foundry, Palantir AIP, Python, FastAPI, PySpark, SQL, OpenAI, Anthropic Claude, Gemini, Mistral, LangChain, LlamaIndex, RAG, Vector Databases, AWS, Azure, Terraform, REST APIs, OAuth2/OIDC, JWT, Docker, CI/CD, LLMOps/MLOps.

Software Engineer AI *Client: CVS*

Dec 2023 – Sept 2024
Remote, USA

- Designed and developed enterprise healthcare applications using **Python, FastAPI, Flask, REST APIs, AWS, SQL, and microservices**, translating complex business requirements into scalable backend services and cloud-native workflows.
- Built **GenAI, LLM, RAG, and agentic AI prototypes** using OpenAI, LangChain, embeddings, vector databases, prompt engineering, semantic retrieval, models, agents, and structured enterprise healthcare data.

- Developed Python-based APIs and microservices integrating **model providers, healthcare data sources, workflow services, third-party APIs, and cloud services** through secure, reusable enterprise integration patterns.
- Engineered cloud-native solutions using **AWS Lambda, S3, IAM, CloudWatch, Docker, and serverless architectures**, supporting scalable event-driven workflows, asynchronous processing, and enterprise application integrations.
- Implemented secure application and API integrations using **OAuth2/OIDC, JWT, RBAC, REST APIs, Swagger/OpenAPI, authentication, authorization, security, privacy, and enterprise data-handling controls**.
- Applied **MLOps/LLMOps, model evaluation, prompt management, monitoring, logging, CI/CD, Jenkins, GitLab, and Git** practices to improve deployment repeatability, observability, and operational readiness.
- Collaborated with product teams, Solution Architects, and engineering stakeholders through **technical discovery, solution design, feasibility analysis, architecture discussions, implementation planning, technical documentation, and production deployment**.

Environments: Python, FastAPI, Flask, OpenAI, LangChain, RAG, Embeddings, Vector Databases, Agentic AI, AWS Lambda, S3, IAM, CloudWatch, REST APIs, Microservices, OAuth2/OIDC, JWT, Docker, Jenkins, GitLab CI/CD, SQL, MLOps/LLMOps.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and developed enterprise backend applications using **Python, FastAPI, Flask, REST APIs, SQL, microservices, and object-oriented programming** for scalable business applications and distributed enterprise systems.
- Built reusable Python microservices implementing **business logic, data transformations, workflow orchestration, API integrations, asynchronous processing, and enterprise application services** across complex operational workflows.
- Engineered data-processing solutions using **Python, Pandas, PySpark, SQL, ETL/ELT pipelines, PostgreSQL, MySQL, and Oracle** to process, transform, and integrate large transactional and operational datasets.
- Integrated internal and external enterprise platforms through **REST APIs, JSON, Swagger/OpenAPI, authentication, reusable API clients, data sources, workflow tools, and event-driven architecture patterns**.
- Contributed to cloud and application modernization initiatives using **Docker, cloud-native architecture, CI/CD, Jenkins, Git, and automated deployment workflows**, improving portability and deployment consistency across environments.
- Applied software engineering practices involving **modular architecture, reusable libraries, exception handling, structured logging, security, scalability, maintainability, performance optimization, and technical documentation**.
- Collaborated with architects, product teams, and engineering stakeholders through **technical discovery, solution design, feasibility assessment, sprint planning, dependency analysis, implementation, release delivery, and production support**.

Environments: Python, FastAPI, Flask, Pandas, PySpark, REST APIs, Microservices, SQL, PostgreSQL, MySQL, Oracle, ETL/ELT, Docker, Jenkins, Git, CI/CD, Cloud Computing, Distributed Systems, Event-Driven Architecture, Agile/Scrum.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed backend services and enterprise applications using **Python, Flask, FastAPI, REST APIs, SQL, and microservices architecture** across distributed and client-facing application environments.
- Built reusable Python modules for **business logic, data transformation, workflow processing, API communication, configuration management, logging, and exception handling**, improving modularity and maintainability.
- Developed RESTful APIs connecting **web applications, backend services, relational databases, data sources, workflow components, and third-party enterprise platforms** through JSON-based integration patterns.

- Designed data-access and processing components using **Python, PostgreSQL, MySQL, SQL, batch processing, scheduled workflows, and high-volume transactional datasets** for enterprise business operations.
- Contributed to **microservices, distributed systems, event-driven architecture, API contracts, service boundaries, authentication, authorization, security, and enterprise integration** patterns across application platforms.
- Implemented reliable software delivery practices using **Git, CI/CD, Linux, Docker, source versioning, application builds, deployment workflows, logging, and environment configuration**.
- Collaborated with architects, senior engineers, business analysts, and stakeholders through **requirements discovery, technical design, implementation planning, development, technical documentation, deployment, handoff, and production support**.

Environments: Python, Flask, FastAPI, REST APIs, Microservices, SQL, PostgreSQL, MySQL, JSON, Distributed Systems, Batch Processing, Event-Driven Architecture, Git, CI/CD, Docker, Linux, Cloud Computing, Agile/Scrum.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.